Jingyun Yang

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Research Interests

I'm enthusiastic about building learning frameworks that empower autonomous agents with the capability to perceive the world via *computer vision*, develop representation of entities and tasks via *program synthesis*, and interact with the world via *predicative planning* or *reinforcement learning*.

Education

University of Southern California — Los Angeles, CA
Bachelor of Science in Computer Science · Minor in Mathematics · GPA: 4.0 — *August 2016 – May 2020*

Publications

1. Youngwoon Lee, **Jingyun Yang**, Joseph J. Lim, "Learning to Coordinate Manipulation Skills via Skill Behavior Diversification", International Conference on Learning Representations (ICLR), 2020.
2. Karl Pertsch, Oleh Rybkin, **Jingyun Yang**, Konstantinos Derpanis, Kostas Daniilidis, Joseph J. Lim, Andrew Jaegle, "KeyIn: Keyframing for Visual Planning", Learning for Dynamics and Control (L4DC), 2020.
3. Kien Nguyen, **Jingyun Yang**, Yijun Lin, Jianfa Lin, Yao-Yi Chiang, Cyrus Shahabi, "Los Angeles Metro Bus Data Analysis Using GPS Trajectory and Schedule Data (Demo Paper)", ACM SIGSPATIAL, 2018.

Experience

USC Cognitive Learning for Vision and Robotics (CLVR) Lab — Los Angeles, CA
Research Assistant · Supervisor: Joseph J. Lim — *August 2018 – May 2020*

- **Neural Program Synthesis from Visual Demonstrations:** Developing a deep learning model that synthesizes behavior programs for realistic videos where behavior segments are temporally extended.
- **Learning Coordinated Tasks from Rewards:** Designed, implemented, and tested a hierarchical reinforcement learning model that tackles complex tasks of multiple agents by learning composable primitive skills and coordination of the skills. Paper accepted in ICLR 2020.
- **Video Prediction with Temporal Hierarchy:** Developed experiments and designed downstream planning algorithm for a model that learns to discover informative frames in a future video sequence and represent the video via its keyframes.

USC Integrated Media Systems Center / USC Spatial Sciences Institute — Los Angeles, CA
Research Assistant · Supervisor: Yao-Yi Chiang — *August 2016 – August 2018*

- **Neural Style Transfer:** Experimented with deep learning methods that transform a map dataset with arbitrary styles to a dataset with target styles required for text detection training.
- **Text Detection:** Experimented with various state-of-the-art text detection models on historical map datasets.
- **Timeseries Data Analytics:** Analyzed dynamic and large-scale transportation sensor data sent by buses in LA.

Bloomberg L.P. — New York, NY
Software Engineering Intern · Supervisor: Luke Shen — *May 2018 – August 2018*

- **Backend Software Development:** Migrated core functionalities of the Company Financials Core Algorithms Library to Spark in Scala.

USC Information Sciences Institute — Los Angeles, CA
Research Assistant · Supervisor: Yolanda Gil — *May 2017 – August 2017*

- **Data Visualization:** Created a module that automatically generates visualizations for either an arbitrary dataset or a group of datasets in the workflow system.

Awards and Honors

- USC Provost's Undergraduate Research Fellowship — *May, August 2019*
- Tau Beta Pi the Engineering Honor Society — *April 2018*
- USC Academic Achievement Award — *January, August 2017*

Teaching

- CSCI 170 – Discrete Methods in Computer Science, Teaching Assistant, USC — *Spring, Fall 2019*
- CSCI 270 – Introduction to Algorithms and Theory of Computing, Teaching Assistant, USC — *Fall 2018*